Fully 22/4/2026

DATE

EXP - NO 1

S.R NO	Description	Value A	
1	weight of Empty cousible	A gm	21.74
2	weight of crucible and coal and before heating at 110° c	B gm	22.25
3	weight of coal	2gm (B-A)	0.51
4	weight of crucible + coal (after heating)	C gm	22.21 gm
5	loss in weight (Due to removal of moisture	B-C gm	0.04 gm
6	wt of crucible and coal (After heating at 925°c)	C gm	22.24 gm
7	wt of crucible and coal (After heating at 925 °c)	D gm	21.97 gm
8	loss in wt (Due to Removal of volatile matter)	C - D gm	0 - 27 gm
9	wt. of crucible & ash (after heating at 750°c for 30 min)	E - gm	21.73 gm
10	weight of Ash	E-A gm	0.01 gm

1] % moisture = $\frac{\text{loss in weight}}{\text{weight of coal}} \times 100$

$= \frac{0.04 \times 100}{0.5}$

% moisture = 8%.

2] % volatile matter = $\frac{\text{loss in weight}}{\text{weight of coal}} \times 100$

% volatile matter = 46%.

3] % Ash = $\frac{\text{weight of Ash}}{\text{weight of coal}} \times 100$

$= \frac{0.01 \times 100}{0.5}$

% Ash = 2%.

4] fixed carbon

FC = 100 - (m + vm + Ash)

= 100 - (56)

FC = 44%.

Result

Sr. No	Description	%
1	moisture	8%
2	Voltalile matter	46%
3	Ash content	2%
4	fixed carbon 100 - (% moisture + % volatile matter + % Ash content)	44%